AQ8.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Choose the correct options.

☐ Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be a linear transformation, where R^2 R2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$ Tu · Tv = u · v.

☐ Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be an orthogonal linear transformation, where R^2 R2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$ Tu · Tv = u · v.

☐ Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be a linear transformation, where R^2 R2 is the inner product space with respect to the inner product given by $< a, b > = 2a_1 b_1 + 5a_2 b_2$ < a, b >= 2a1b1 + 5a2b2. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$ ⟨Tu, Tv⟩ = ⟨u, v⟩.

☐ Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be an orthogonal linear transformation, where R^2 R2 is the inner product space with respect to the inner product given by $< a, b > = 2a_1 b_1 + 5a_2 b_2$ < a, b >= 2a1b1 + 5a2b2. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$ ⟨Tu, Tv⟩ = ⟨u, v⟩.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be an orthogonal linear transformation, where R^2 R2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$ Tu · Tv = u · v.
Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be an orthogonal linear transformation, where R^2 R2 is the inner product space with respect to the inner product given by $< a, b > = 2a_1 b_1 + 5a_2 b_2$ < a, b >= 2a1b1 + 5a2b2. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$ ⟨Tu, Tv⟩ = ⟨u, v⟩.

1 point
If A A is the matrix representation of an orthogonal transformation $T: R^n \rightarrow R^n$ T : Rn → Rn, then choose the set of correct statements.

☐ $AA^T = I$ AAT = I

☐ $A = A^T$ A = AT

☐ AA^T AAT is an upper triangular matrix

☐ AA^T AAT is a lower triangular matrix

☐ $A^T = A^{-1}$ AT = A−1

☐ $A^T = -A^{-1}$ AT = −A−1

☐ A^{-1} A−1 does not exist

☐ A^{-1} A−1 is also an orthogonal matrix

No, the answer is incorrect.
Score: 0

Accepted Answers:
$AA^T = I$AAT = I
AA^T AAT is an upper triangular matrix
AA^T AAT is a lower triangular matrix
$A^T = A^{-1}$AT = A−1
A^{-1} A−1 is also an orthogonal matrix

1 point

If AA is the matrix representation of an orthogonal transformation $T: R^n \rightarrow R^n$T : Rn → Rn, then choose the set of correct statements.

☐ The column vectors of AA are orthogonal but not orthonormal.

☐ The column vectors of AA are orthonormal.

☐ The row vectors of AA are orthogonal but not orthonormal.

☐ The row vectors of AA are orthonormal.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The column vectors of AA are orthonormal.
The row vectors of AA are orthonormal.

Level 2

1 point

Consider the orthogonal set $B = \{(1,1,1),(-1,1,0)\}$B = {(1, 1, 1), (−1, 1, 0)} in R^3 R3. Find a third vector in R^3 R3 which is orthogonal to both the vectors in BB.

○ $(1,1,-2)$(1, 1, −2)

○ $(1,-1,2)$(1, −1, 2)

○ $(-1,1,-2)$(−1, 1, −2)

○ $(-1,-1,-2)$(−1, −1, −2)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(1,1,-2)$(1, 1, −2)

1 point

Consider the orthogonal set $B = \{(1,1,1),(-1,1,0)\}$B = {(1, 1, 1), (−1, 1, 0)} and vector obtained in question 4 which is orthogonal to the set BB. Form the columns of an orthogonal matrix UU by scaling these vectors appropriately. Choose the correct option.

○ $U = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$U = [1 −1 1; 1 1 1; 1 0 −2]

○ $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$ [; ;]

$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix} U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$

$U = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix} U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$

$$\begin{bmatrix} & \\ & \\ -2 & \end{bmatrix}$$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix} \quad U = \begin{bmatrix} & & \\ & & \\ & 3 & \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{1} & \sqrt{3} \\ \sqrt{1} & \sqrt{3} \\ \sqrt{1} & \sqrt{2} \\ \sqrt{-1} & \sqrt{2} \\ \sqrt{10} & \sqrt{6} \\ \sqrt{1} & \sqrt{6} \\ \sqrt{1} & \sqrt{6} \\ \sqrt{} & \\ -2 & \end{bmatrix}$$

1 point

Consider a vector $v = (3, 0, 4)$v = (3, 0, 4) and UU from Question 5. Choose the correct option using the dot product as the standard inner product.

- ◯ $\| v \| = \| Uv \| \|v\| = \|Uv\|$
- ◯ $\| v \| \neq \| Uv \| \|v\| = \|Uv\|$
- ◯ $2 \| v \| = \| Uv \| 2\|v\| = \|Uv\|$
- ◯ $\| v \| = 2 \| Uv \| \|v\| = 2\|Uv\|$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\| v \| = \| Uv \| \|v\| = \|Uv\|$

1 point

Let the matrix representation of a linear transformation $T: R^3 \rightarrow R^3$ T : R3 → R3 be $U = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$U = $\begin{bmatrix} \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{-1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{bmatrix}$. Let $B = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}$B = $\begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}$. Choose the set of correct statements.

- [ ] TT is an orthogonal transformation.
- [ ] TT is not an orthogonal transformation.
- [ ] BB is a real symmetric matrix.
- [ ] $U^{-1}BU$U−1BU is a diagonal matrix.
- [ ] $U^{-1}BU$U−1BU is a scalar matrix.
- [ ] $U^{-1}BU$U−1BU is an upper triangular matrix.
- [ ] $U^{-1}BU$U−1BU is a lower triangular matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

TT is an orthogonal transformation.

BB is a real symmetric matrix.

$U^{-1}BU$U−1BU is a diagonal matrix.